

What is a ReplicaSet?

Simple Definition

A ReplicaSet ensures that a specified number of identical Pods are always running.

Beginner Definition

A ReplicaSet is a Kubernetes object that automatically creates and maintains multiple copies (replicas) of a Pod.

Why Do We Need a ReplicaSet?

Imagine your website has **only one Pod**.

Website

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Pod 1

If Pod 1 crashes:

Website

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✗ Pod 1 Crashed

Your website goes down.

A ReplicaSet solves this problem.

If you tell Kubernetes:

"I always want 3 Pods running."

Then:

ReplicaSet

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|— Pod 1

└─ Pod 2

└─ Pod 3

If Pod 2 crashes:

ReplicaSet

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└─ Pod 1

└─ ❌ Pod 2

└─ Pod 3

ReplicaSet automatically creates a new Pod:

ReplicaSet

|

└─ Pod 1

└─ Pod 3

└─ New Pod 4

👉 This is called **Self-Healing**.

ReplicaSet YAML Explained

1. API Version

apiVersion: apps/v1

Meaning

Specifies the API version.

Pods use:

apiVersion: v1

ReplicaSets use:

apiVersion: apps/v1

Remember:

- Pod → v1
 - ReplicaSet → apps/v1
-

2. Kind

kind: ReplicaSet

Meaning

Tells Kubernetes:

"Create a ReplicaSet."

Examples:

kind: Pod

kind: Service

kind: ReplicaSet

kind: Deployment

3. Metadata

metadata:

Stores information about the ReplicaSet.

Name

name: myapp-replicaset

This is the ReplicaSet name.

Example:

ReplicaSet Name

myapp-replicaset

Labels

labels:

app: myapp

Labels identify the ReplicaSet.

Think of them as a name tag.

4. Spec

spec:

Spec tells Kubernetes:

"How should this ReplicaSet work?"

5. Replicas ★ (Most Important)

replicas: 3

Meaning

Always keep **3 Pods running**.

If one Pod fails:

ReplicaSet creates another one automatically.

Example:

Desired Pods = 3

Running

Pod 1

Pod 2

Pod 3

If Pod 2 crashes:

ReplicaSet

Pod 1

Pod 3

Creates

New Pod

Again:

Pod 1

Pod 3

Pod 4

Total = 3 Pods

6. Selector

selector:

Meaning

Selector tells ReplicaSet:

"Which Pods should I manage?"

Match Labels

matchLabels:

app: myapp

ReplicaSet looks for Pods with:

app: myapp

Only those Pods are managed.

Why Must Labels Match?

Notice these two sections:

selector:

matchLabels:

app: myapp

and

template:

metadata:

labels:

app: myapp

Both have:

app: myapp

This is very important.

ReplicaSet uses the selector to find Pods with the matching label.

If they don't match, ReplicaSet won't manage those Pods.

7. Template ★

template:

Meaning

This is the **blueprint** for every Pod the ReplicaSet creates.

Think of it as a photocopy machine.

Every Pod created by the ReplicaSet follows this template.

Template Metadata

metadata:

labels:

app: myapp

Every Pod created will have:

app: myapp

Template Spec

spec:

Describes how each Pod should run.

Containers

containers:

Each Pod contains:

- name: nginx

Container name = nginx

Image

image: nginx

Use the nginx Docker image.

Every Pod created will run an nginx container.

Complete Flow

ReplicaSet

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Desired Replicas = 3



Creates 3 Pods



Each Pod runs an nginx Container

If one Pod crashes:

ReplicaSet



Detects Missing Pod



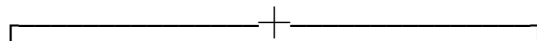
Creates New Pod

Easy Diagram

ReplicaSet



replicas = 3



Pod 1



Pod 2



Pod 3



nginx nginx nginx

If Pod 2 crashes:

ReplicaSet

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Creates New Pod



Pod 4

Complete YAML with Explanation

```
apiVersion: apps/v1      # API version for ReplicaSet
```

```
kind: ReplicaSet         # Create a ReplicaSet
```

```
metadata:
```

```
  name: myapp-replicaset  # ReplicaSet name
```

```
labels:
```

```
  app: myapp              # Label for ReplicaSet
```

```
spec:
```

```
  replicas: 3             # Keep 3 Pods running
```

```
selector:
```

matchLabels:

app: myapp # Manage Pods with this label

template: # Blueprint for new Pods

metadata:

labels:

app: myapp # Label assigned to created Pods

spec:

containers:

- name: nginx # Container name

image: nginx # Docker image

Pod vs ReplicaSet

Pod	ReplicaSet
Creates one Pod	Creates multiple Pods
No self-healing	Self-healing if a Pod fails
Manual scaling	Automatic maintenance of desired Pod count
Good for learning	Used to keep applications available

KCNA Revision Points ★

- apiVersion: apps/v1 → API version for ReplicaSet.

- kind: ReplicaSet → Create a ReplicaSet.
 - metadata → Information about the ReplicaSet.
 - replicas: 3 → Always maintain 3 Pods.
 - selector → Finds Pods to manage.
 - matchLabels → Must match the Pod template labels.
 - template → Blueprint used to create Pods.
 - containers → Defines the container(s) inside each Pod.
 - image: nginx → Uses the nginx Docker image.
 - **ReplicaSet automatically recreates Pods if they fail (self-healing).**
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Easy Memory Trick

ReplicaSet

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"I want 3 Pods"

|



Creates Pods

|



One Pod crashes

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ReplicaSet creates a new Pod automatically

One-line Formula:

ReplicaSet = Desired Number of Pods + Self-Healing

This is the main purpose of a ReplicaSet in Kubernetes.